

Ali Alyaseri

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EDUCATION

University of Toronto

BASc Computer Engineering + PEY Co-op

Minors in Engineering Business and AI; Certificate in Cybersecurity

Toronto, ON

Expected Dec 2027

- **Relevant Courses:** Operating Systems (virtual memory, threads), Computer Networks, Computer Security, Computer Systems Programming, Programming Languages, Software Engineering, Databases, Control Systems, Foundations of Cybersecurity, Applied Deep Learning (PyTorch)
- **Awards:** Jeffrey Skoll Scholarship; Dean's List; **1st of 120 teams** in Microchip-sponsored embedded competition; MakeUofT: **Best Use of Presage Heart Webcam** and **2nd, Qualcomm Uno Q**

EXPERIENCE

Quanta Technology / Danovo Energy Solutions

Ontario

Software Engineering Intern, Power Systems Digital Transformation

May 2024–Present

- Optimized C++/C# grid-simulation code in **APA (formerly CAPE)**, refactoring data access and parallel execution to improve performance by **28%**.
- Automated batch power-system studies and result validation with **Python**, saving an estimated **8–10 hours/week** of manual analysis and reporting.
- Built a **React/TypeScript + Express/MongoDB** license-management platform with identity sync, cost allocation, audit trails, and Docker deployment.
- Standardized **protection-compliance data flow** across React, ASP.NET Core, and C# engines, fixing utility-specific baseline loading and equipment matching with regression tests.
- Refactored an **Oracle-backed settings-comparison engine** to avoid driver type failures on large datasets; added async retrieval and regression tests across input formats.
- Hardened **React/TypeScript** simulation polling and chunked uploads against timeouts, races, and stale-cache failures.

TinyHorse

Ontario

Software Engineer / Python, JavaScript/TypeScript, AWS Lambda

Dec 2023–May 2024

- Cut **Python AWS Lambda** API latency by **25%** through serialization and data-access optimization.
- Built **Next.js** interfaces and AWS Lambda/API Gateway pipelines integrating LLM APIs.

Shearer's

Ontario

IT Administrator / SQL, VPN, network hardware

May 2023–Sep 2023

- Secured VPN/VNC access for **200+ users**; maintained SQL data and network hardware.

PROJECTS

Multi-Agent RTL Debug and Repair System (Capstone) | Python, Verilog, LLM agents

- Building agents to generate tests, **localize RTL failures** from waveforms, suggest fixes, and independently validate patches with lint and simulation on **Verilator-compatible designs**.

Embedded Thermal Tracking System | C, STM32, signal processing | [code](#)

- Won **1st of 120 teams**; wrote STM32 C firmware that expands **8x8 to 128x128** sensor frames, thresholds heat regions, and extracts connected components and centroids under MCU memory limits. Added **Kalman-filtered multi-target tracking** and confidence-gated pan/tilt servo control to smooth noisy detections.

GIS Mapping and Routing | C++, GTK, OpenStreetMap

- Parsed **OpenStreetMap** entities into spatially indexed C++ structures for fast lookup and GTK rendering; implemented **Dijkstra** and **A*** search to route across street graphs.

FPGA Real-Time Co-op Game | Embedded C, Nios II, Verilog | [code](#)

- Programmed game state, collision handling, and **VGA rendering** in C on a **Nios II soft processor**; coordinated PS/2 input and display updates through **interrupts and memory-mapped I/O** to keep the gameplay loop responsive within FPGA timing and resource limits. Simulated Verilog timing modules in ModelSim before integration.

SMILE Dental X-ray Detection | Python, PyTorch, YOLO, CBAM | [code](#)

- Unified CSV, YOLO, and mask labels for roughly **975 X-rays / 5,355 cavities**; trained a **YOLO + CBAM** detector with mosaic and MixUp augmentation to localize subtle lesions.

TECHNICAL SKILLS

Languages: C, C++, C#, Python, Verilog, ARM assembly, SQL, JavaScript/TypeScript

Software and cloud: React, ASP.NET Core, WPF/XAML, Next.js, Node.js/Express, AWS EC2 and Lambda, REST APIs

Data and engineering: SQL Server, Oracle/Insights, MongoDB, APA (formerly CAPE), ASPEN OneLiner, PyTorch

Tools: ModelSim, Quartus, AMD and Altera FPGAs, Nios II, STM32, LTspice, MATLAB, Git, Linux, Docker, VS Code, Jira, Figma, Valgrind, perf

Systems and concepts: digital logic, FPGA development, memory-mapped I/O, multithreading, performance profiling, network/packet optimization, API development, AI-assisted tooling (Cursor, Codex, MCP)